Attention with a Calibrated Off Switch: Causal FIR Key Smoothing, the Score-Shift Pathology, and Selective Silence in Rectified Attention (`pn_fir_tau`)

Maxim Zhukov*
Independent Research

Reproducibility Package: `src/`, `experiments/exp1..6`, `exp4b`, `exp4c`, `exp4e`, `exp4f`, `exp4g`, `exp4h`, `results/`, `run_all_and_build_figures.py`

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Abstract

Softmax attention cannot say "nothing found": a simplex-constrained head must distribute 1.000 units of context weight even when no key matches the query, and pre-trained models pay with attention sinks, massive outliers that degrade low-bit quantization [13, 2, 4], and RAG-style hallucinations on unanswerable queries. The off-switch mechanism itself is established prior art: ReLA replaces Softmax with ReLU and its heads "attend to nothing" [1]; Softpick scales rectified normalization to 340M–1.8B with a 0% sink rate [2]. The open question is the interaction layer audited here. A parameter-free 7-tap causal FIR low-pass filter ($\sigma = 2.0$) applied strictly to the key stream K lets one layer bind a queried key onto the adjacent value slot, lifting non-bijective single-layer associative recall from $44.5\% \pm 0.9\%$ to $100.0\% \pm 0.0\%$ and removing exact-decoy vulnerability on the legacy benchmark ($\Delta = 0.000$ at decoy noise $\epsilon \in \{0.05, 0.2, 0.5, 1.0\}$). The same smoothing shifts scores downward and starves rectification: fixed-τ=0 power-norm falls to $64.4\% \pm 6.8\%$ with $91.3\% \pm 8.6\%$ negative scores and $44.5\% \pm 23.0\%$ dead rows; Softmax masks this shift (translation-invariant), rectification does not (Proposition 1). One learnable threshold τ per layer (`pn_fir_tau`) restores 100.0% recall and keeps unanswerable rows silent under pure cross-entropy ($\sum_j W_{t,j} = 0.002 \pm 0.002$ vs. 0.100 ± 0.005 for virtual sinks) reaches matched-precision WikiText-2 parity on 2,400 steps ($\leq 0.3\%$ across five variants, single seed). We release fused Triton kernels (no exp(), $O(N)$ attention VRAM) and a full measurement record, including an instrumented retraction of our initial end-to-end latency claim.

1 Introduction: Practical Motivation and Related Work

Why modify the core attention mechanism of a Transformer? In practical sequence modeling, standard scaled dot-product Softmax attention [7] forces practitioners to pay a steep architectural and computational tax for two basic operations: **(A) reading the value token that follows a matched key in context** (in-context learning and associative lookup), and **(B) doing nothing**

*ORCID: 0009-0005-9238-3778

when a specialized attention head finds no relevant match in the context (residual-stream preservation in RAG, tool use, and memory lookup).

1.1 Practical Problem 1: Compressing Two-Layer Induction Heads into a Single Layer (`sm_fir` / `pn_fir_tau`)

Associative retrieval—locating a context key k^* that matches a query and routing the value v^* stored at an adjacent position j^*+1 (`[... key A, value B ... query A] -> B`)—is the foundational mechanism of in-context learning in Large Language Models (LLMs), enabling models to copy variable names, track entity attributes, and extract facts from prompts [6]. In a standard Transformer, however, **a single token-level attention layer is structurally incapable of content-addressable key-value induction**: dot-product similarity $S_{t,j} = \langle Q_t, K_j \rangle / \sqrt{d_k}$ matches the query Q_t (encoding k^*) against the key vector K_{j^*} at the key slot j^*, whereas the target value token v^* resides at the adjacent slot j^*+1, whose unfiltered key vector $K_{j^*+1} = W_K E(v^*)$ carries zero information about k^* whenever keys and values are independently paired. As proved by Olsson et al. [6], a standard Transformer must dedicate **at least two sequential layers** to solve this task: Layer 1 (*previous-token head*) shifts token j^* into the residual stream of position j^*+1, and only in Layer 2 (*induction head*) can attention match Q_t against the composite representation at j^*+1. In compact or edge Transformers (4–12 layers), spending early layers and residual-stream bandwidth just to shift adjacent tokens is an expensive waste of depth.

Applying a parameter-free causal Finite Impulse Response (FIR) low-pass filter strictly to the key stream K along the sequence dimension (the convolutional ancestors of this move are RWKV's learnable 2-tap token shift [5] on the R/K/V streams and the searched depthwise convolutions of *Primer* [8]; ours is a fixed zero-parameter $K_f = 7$ half-Gaussian low-pass strictly on K) blends the representation of k^* at position j^* forward into position j^*+1:

$$\tilde{K}_{j^*+1} = h_0 K_{j^*+1} + h_1 K_{j^*} + \sum_{m=2}^{K_f-1} h_m K_{j^*+1-m}. \tag{1}$$

By Eq. (1), $\tilde{K}_{j^*+1}$ now contains a weighted component $h_1 K(k^*)$ while both the query Q_t and the value stream $V_{j^*+1} = W_V E(v^*)$ remain unfiltered, so **a single attention head querying for k^* can attend directly to the value slot j^*+1 and read out v^***—lifting single-layer non-bijective recall from 44.5% (`sm`) to 100.0% ± 0.0% (`sm_fir` and `pn_fir_tau`) at a latency cost of just +1.5%–2.0% in `sm_fir`.

1.2 Practical Problem 2: Preventing RAG Context Hallucinations and Eliminating Attention-Sink Outliers (`pn_fir_tau`)

In a multi-layer Transformer, the residual stream acts as a shared communication bus ($y_t = x_t + \sum_{j \le t} W_{t,j} V_j$). Most specialized attention heads (e.g., heads looking up an entity attribute or an answer in a retrieved RAG passage) are only relevant on a small fraction of tokens; when a retrieved passage does not contain the queried fact (**unanswerable query**), the head *should stay completely silent* ($\sum_{j \le t} W_{t,j} = 0$) so that the existing residual representation x_t is preserved uncorrupted. Because standard Softmax enforces $\sum_{j=0}^{t} W_{t,j} \equiv 1$ for every query t, it causes two major practical failures (recently modelled as Softmax's two missing primitives, abstention and noise filtering [3]): (1)

RAG context hallucinations on unanswerable queries, where the head is forced to overwrite the residual state x_t with a weighted average of irrelevant context values V_j; and (2) **Attention Sinks and quantization outliers** [13] [14], where pre-trained LLMs turn the initial `BOS` token ($t = 0$) into a "trash bin" by inflating its hidden activations into massive numerical outliers [4] ($W_V x_0 \approx 0$) that degrade INT8/INT4 post-training quantization [13] (a failure mode rectified normalizers remove at 340M–1.8B scale with a low-bit win [2]) and cause perplexity collapse in streaming KV caches whenever the initial sink tokens are evicted [14].

In the sparse and quiet attention literature, two families have been proposed:

1. **Simplex-projected sparse normalizers (`Sparsemax` [9], α-`Entmax` [10], [11]):** For $\alpha > 1$ and $p = \frac{1}{\alpha-1}$, α-Entmax computes $W_{t,j}^{\text{entmax}}(s) = \max(s_j - \tau^*(s), 0)^p$, where $\tau^*(s)$ is the unique Lagrange multiplier solving $\sum_{j=0}^{t} \max(s_j - \tau^*(s), 0)^p = 1$. Because $\tau^*(s + c\mathbf{1}) = \tau^*(s) + c$, Sparsemax ($\alpha = 2$) and Entmax ($\alpha = 1.5, 1.1$) are translation-invariant, but because $\sum_{j=0}^{t} W_{t,j} \equiv 1$ everywhere, they **can never produce an all-zero row** (zero selective silence), and concentrating weight onto the single least-negative mismatching token actually *increases* context hallucination rates (32.4%–34.4% vs. 27.2% for Softmax in Exp 5).

2. **Virtual-sink Softmax (`Softmax_1` [12], Quiet Softmax [13], Attention Sinks [14]):** Appending a virtual sink logit τ yields $W_{t,j} = \frac{\exp(s_j)}{\sum_{m=0}^{t} \exp(s_m) + \exp(\tau)}$, breaking shift-invariance so $\sum_{j=0}^{t} W_{t,j} < 1$, but retaining strictly positive exponential tails $\exp(s_j) > 0$ across all context positions ($\sum_j W_j = 0.100 \pm 0.005$ leakage on absent keys in Exp 5).

3. **Rectified (Sub-Simplex) Normalizers — established prior art (ReLA [1], Softpick [2]):** ReLA [1] replaces Softmax with ReLU, observes sparsity without projection, reports heads that "attend to nothing," and stabilises training via LayerNorm initialisation or gating; Softpick [2] trains a rectified, deliberately non-sum-to-one normalizer at 340M–1.8B with a 0% sink rate and better low-bit quantized accuracy than Softmax. `pn_fir_tau` differs from this line in one falsifiable respect: it couples a per-layer learnable threshold τ with a peak-driven silence gate, and the threshold exists to calibrate the score shift that causal key smoothing imposes — not merely as a stability device.

Given that map, the open problem addressed here is not whether attention can be silent — prior art answers that — but whether a rectified normalizer survives causal key smoothing. Our audit (Section 4) uncovers a striking paradox when training is dominated by positive retrieval examples: adding causal FIR on K to Softmax (`sm_fir`) lifts clean associative recall to $100.0\% \pm 0.0\%$, whereas adding the exact same causal FIR filter on K to uncalibrated rectified power-normalization (`pn_fir`, fixed $\tau = 0$) degrades recall to $64.4\% \pm 6.8\%$ (Exp 1) and $88.8\% \pm 9.2\%$ (Exp 2), while row-mean centering (`pn_fir_c`) makes row mass dependent on within-row dispersion rather than absolute score level.

Scope. This is a controlled mechanistic study of how causal FIR smoothing interacts with sub-simplex normalization; the 1.8M-parameter language model and the synthetic recall tasks are deliberately small sandboxes that isolate the mechanism, not evidence of LLM-scale behaviour. We frame the contribution as a calibrated diagnostic and tool, and treat scale-up as future work.

2 Mathematical Formulation and Score-Shift Pathology

Let $X \in \mathbb{R}^{B\times T\times D}$ be the input sequence with model dimension D, H heads, and head dimension $d_k = D/H$. Linear projections $Q = XW_Q$, $K = XW_K$, and $V = XW_V$ have shape $\mathbb{R}^{B\times H\times T\times d_k}$.

2.1 Causal FIR Low-Pass Filtering Strictly on the Key Stream K

Non-causal FFT filtering (`sm_fft`, `pn_fft`, `qe_pn_fft`) multiplies the full-sequence spectrum by a Gaussian mask, which leaks future tokens into past keys (past_key_leak = 0.785). To guarantee strict autoregressive causality (past_key_leak = 0.0), we apply a causal depthwise 1D FIR filter of length $K_f = 7$ exclusively to the key stream K, with half-Gaussian weights using $\sigma = \max((K_f - 1)/3.0, 0.5) = (7-1)/3.0 = 2.0$ (`causal_gaussian_taps` in `src/pnfirtau_attention.py`; Fig. 2):

$$h_m = \frac{\exp\left(-\frac{1}{2}(m/\sigma)^2\right)}{\sum_{r=0}^{K_f-1}\exp\left(-\frac{1}{2}(r/\sigma)^2\right)}, \quad m \in \{0,\dots,6\},$$
$$h = [0.3329,\ 0.2938,\ 0.2019,\ 0.1081,\ 0.0451,\ 0.0146,\ 0.0037]. \tag{2}$$

With causal left-padding of $K_f - 1 = 6$ zeros using the taps in Eq. (2), $\tilde{K}_t = \sum_{m=0}^{\min(t,6)} h_m K_{t-m}$ depends exclusively on indices $\le t$, while both Q and V remain unfiltered (Fig. 1), yielding scores $S_{t,j} = \langle Q_t, \tilde{K}_j\rangle/\sqrt{d_k}$ for $j \le t$.

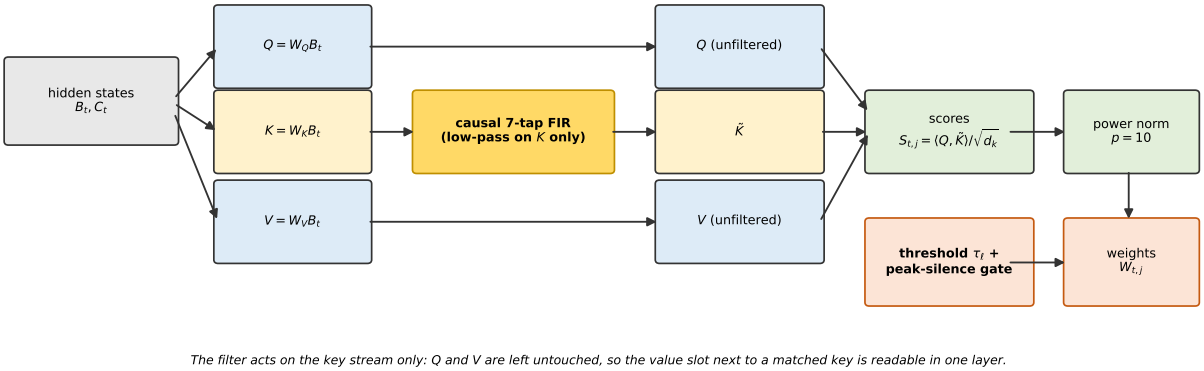


Figure 1: Schematic of the rectified attention head architecture. The causal FIR low-pass filter acts on the key stream only; the query and value streams are left untouched, so a matched key can be read at the adjacent value slot within a single layer. Source: `paper_en.tex` and `src/pnfirtau_attention.py`.

2.2 Simplex vs. Sub-Simplex Normalizers, Shift-Ray Invariance, and Score-Shift Pathology

Proposition 1 (Mass Constancy Along Uniform Shift Rays)**.** *Let $\Phi : \mathbb{R}^n \to [0,1]^n$ be any attention row normalizer with total row mass $M(s) = \sum_{j=1}^n \Phi_j(s) \in [0,1]$.*

1. ***(Simplex normalizers):*** *If Φ maps onto the probability simplex ($M(s) \equiv 1$ for all $s \in \mathbb{R}^n$, as in Softmax, Sparsemax, and α-Entmax), then $M(s)$ is identically 1 everywhere and selective silence ($M(s) \to 0$) is impossible on any score vector $s \in \mathbb{R}^n$.*

2. ***(Shift-invariant normalizers along uniform shift rays):*** *If Φ is translation-invariant ($\Phi(s + c\mathbf{1}) = \Phi(s)$ for all $s \in \mathbb{R}^n, c \in \mathbb{R}$), then its total row mass is constant along every*

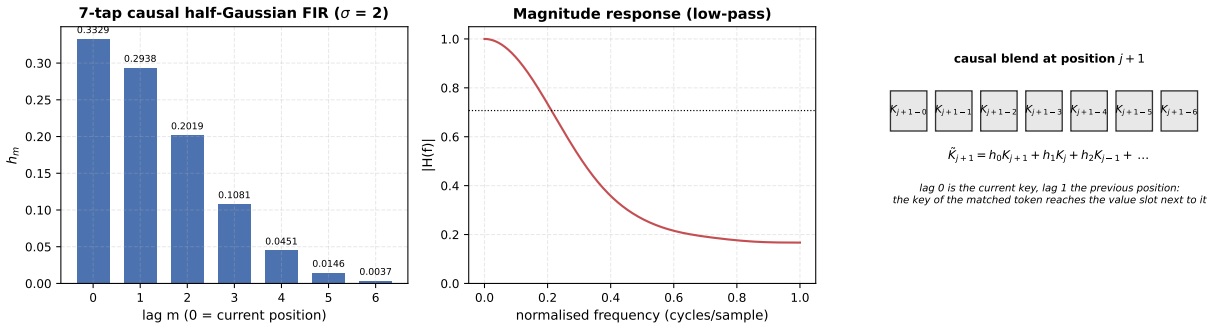


Figure 2: Diagram of the 7-tap causal FIR low-pass filter. Left: the tap coefficients, computed by the production function `causal_gaussian_taps` and normalised to sum to one. Centre: the magnitude response of the same taps. Right: how the causal blend mixes the current key with the preceding positions. The computed taps coincide with the coefficients printed in the paper. Source: `src/pnfirtau_attention.py` and `paper_en.tex`.

> *uniform shift ray* $\{s - c\mathbf{1} : c \in \mathbb{R}\}$*, i.e.,* $M(s - c\mathbf{1}) = M(s)$ *for all* $c \in \mathbb{R}$*. Consequently,* Φ *cannot transition from active retrieval* ($M(s) \approx 1$) *to selective silence* ($M(s - c\mathbf{1}) \to 0$ *as* $c \to +\infty$) *when all scores in a row are shifted downward by a common offset* c*.*

Why distinguish uniform shift rays from within-row dispersion (`pn_fir_c`)? Proposition 1(2) characterizes behavior along *uniform shift rays* $s \mapsto s - c\mathbf{1}$ (where within-row differences $s_j - s_m$ are preserved). A translation-invariant sub-simplex normalizer is not required to have constant mass across *all* $s \in \mathbb{R}^n$: in row-mean-centered power-normalization (`pn_fir_c`, $S'_{t,j} = S_{t,j} - \mu_t$ with peak gate $g_t = \frac{\max_j (S_{t,j}-\mu_t)_+}{\max_j (S_{t,j}-\mu_t)_+ + \gamma}$), $M(S_t) = g_t$ is invariant to uniform shifts $S_t \mapsto S_t - c\mathbf{1}$ because $\max_j S_{t,j} - \mu_t$ is shift-invariant, yet $M(S_t)$ depends on the **within-row score dispersion** $\max_j S_{t,j} - \mu_t$ (and thus on query norm $\|Q_t\|_2$). This explains all three empirical regimes of `pn_fir_c`: (a) without a peak gate ($\gamma = \text{None}$, Exp 2 Table 2), subtracting μ_t guarantees $\max_j (S_{t,j} - \mu_t) > 0$ on every non-constant row, forcing $\sum_j W_{t,j} \equiv 1.000$; (b) on uncalibrated noise rows in positive-only training ($\gamma = 0.5$, Exp 1 Table 1 and Exp 3 Table 5), random dot-product variance keeps $\max_j S_{t,j} - \mu_t \gg \gamma$, yielding $\sum W_{\text{noise}} = 0.938 \pm 0.007$ (gap $\leq +0.021$); and (c) when trained end-to-end with 50% unanswerable queries under Cross-Entropy loss ($\gamma = 0.5$, Exp 5 Table 6), `pn_fir_c` shrinks $\|Q_t\|_2$ on unanswerable queries to achieve partial silence ($\sum W_{\text{unans}} = 0.265 \pm 0.045$, gap $= +0.451 \pm 0.032$), but because $\max_j (S_{t,j} - \mu_t) > 0$ still holds on every non-constant row, it leaks $\approx 124\times$ more residual noise than `pn_fir_tau` (0.002 ± 0.002) while degrading answerable override accuracy ($88.5\% \pm 8.9\%$ vs. $99.0\% \pm 0.7\%$).

Because $\|h\|_2^2 = \sum_{m=0}^{6} h_m^2 = 0.2518 < 1$ and each filtered key $\tilde{K}_j$ averages up to 7 adjacent tokens (most of which are negatively aligned filler tokens during positive-only training), the distribution of causal dot-product scores $S_{t,j} = \langle Q_t, \tilde{K}_j \rangle / \sqrt{d_k}$ contracts in variance and shifts negatively (Fig. 3):

- **Softmax (`sm_fir`), Sparsemax, and Entmax are translation-invariant:** Since $\text{softmax}(S_t - c) = \text{softmax}(S_t)$, Softmax is unaffected by negative offsets. In Exp 1, 100.0% of query rows in trained `sm_fir` have strictly negative scores (raw_frac_pos = 0.0%, raw_rows_all_neg = 100.0%), yet `sm_fir` achieves $100.0\% \pm 0.0\%$ accuracy.

- **Rectified power-norm (`pn_fir`) suffers dead-row starvation under positive-only training:** Uncalibrated power-norm rectifies scores at zero, $S^+_{t,j} = \max(S_{t,j}, 0)$. When

$\max_{j \le t} S_{t,j} \le 0$, the entire row is clipped to zero, yielding $\hat{w}_{t,j} = 0$ and exact zero gradients ($\partial \mathcal{L} / \partial S_{t,j} = 0$). In Exp 1, $91.3\% \pm 8.6\%$ of raw scores in `pn_fir` are negative and $44.5\% \pm 23.0\%$ of query rows are completely dead (raw_rows_all_neg $= 44.5\% \pm 23.0\%$ across seeds).

- **Row-mean centering (`pn_fir_c`) trades absolute level for within-row dispersion:** Subtracting the causal row mean $\mu_t = \frac{1}{t+1} \sum_{m=0}^{t} S_{t,m}$ forces roughly half of every noise row (44.4% of entries in Exp 1) above zero, driving noise attention mass to $\sum_j W_j = 0.938 \pm 0.007$ (Exp 1) and 1.000 ± 0.000 (Exp 2).

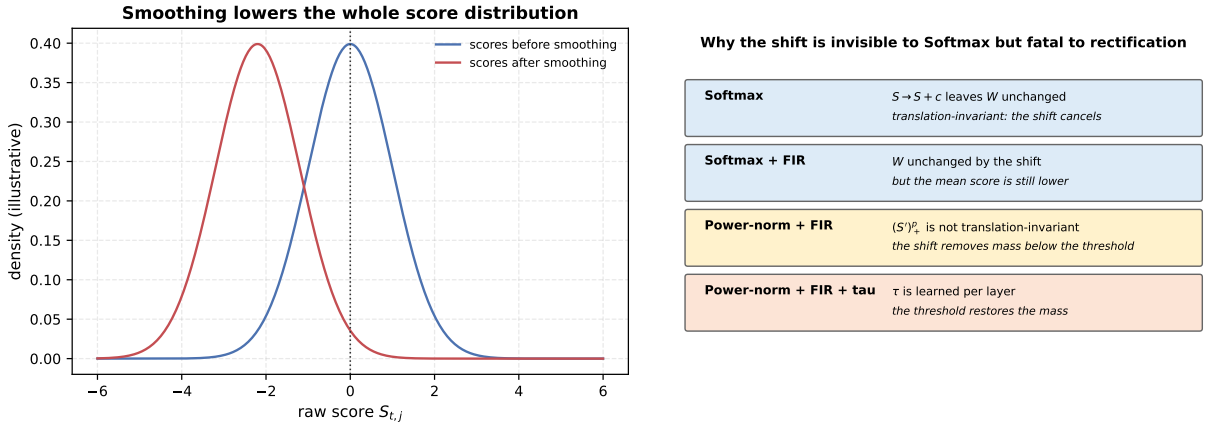


Figure 3: Diagram of the score-shift pathology. Left: smoothing lowers the whole score distribution, which a translation-invariant normaliser cancels but a rectified one does not. Right: the four regimes and why a negative shift is invisible to Softmax yet removes mass under a fixed threshold. The left panel is a schematic illustration of the mechanism; the measured shift is quantified in the text and in the width-ladder figure. Source: `paper_en.tex` (Proposition 1) and `src/pnfirtau_attention.py`.

2.3 The `pn_fir_tau` Formulation and IO-Aware Fused Triton Kernels

To calibrate the global score distribution without restoring within-row shift-invariance, `pn_fir_tau` introduces a single learnable scalar threshold $\tau \in \mathbb{R}$ per attention layer ($\tau_0 = 0.0$):

$$S'_{t,j} = (S_{t,j} - \tau) \cdot \mathbb{I}(j \le t), \qquad S^+_{t,j} = \max(S'_{t,j}, 0), \tag{3}$$

$$\hat{w}_{t,j} = \frac{(S^+_{t,j})^p}{\sum_{m=0}^{t} (S^+_{t,m})^p + \varepsilon}, \qquad g_t = \frac{\max_{0 \le m \le t} S^+_{t,m}}{\max_{0 \le m \le t} S^+_{t,m} + \gamma}, \qquad W_{t,j} = g_t \cdot \hat{w}_{t,j}, \tag{4}$$

where $p = 10.0$, $\varepsilon = 10^{-12}$, and $\gamma = 0.5$ is the peak silence gate constant in Eqs. (3)–(4). The total row weight is $\sum_{j=0}^{t} W_{t,j} = g_t \in [0, 1)$.

In eager PyTorch (`PnFirTauMHA`), evaluating Eqs. (3)–(4) materializes four $[B, H, N, N]$ intermediate tensors in HBM. In `src/triton_pnfirtau.py`, we eliminate this bottleneck via three custom Triton [15] kernel suites: (1) `triton_causal_fir` (zero-copy strided $K_f = 7$ causal FIR on K); (2) `PnFirTauTritonMHA` (`triton_pn_fir_tau_scores`, a single row-fused kernel replacing the 6-op eager score-to-weight chain); and (3) `PnFirTauFlashMHA` (`flash_pn_fir_tau_attention`, an online-tiled $O(N)$-VRAM FlashAttention-2 style kernel with uniform tile size $B_M = B_N = 64$). Crucially, once fused in SRAM registers, $(S'_{t,j})^{10}_+$ is evaluated in 3 register multiplications $(u^2, u^4, u^8 \cdot u^2)$ with

zero transcendental `exp()` calls, accumulates across tiles without online max-rescaling, computes exact analytical gradients in a single KV-parallel backward pass using the identity $D_t \triangleq \sum_{j=0}^{t} dW_{t,j}\hat{w}_{t,j} = \langle dO_t, O_t\rangle / g_t$ with deterministic per-block `FP64` reduction for $\partial\mathcal{L}/\partial\tau$, and includes early-exit tile branches when $\max u_{t,j} \leq 0$ across a 64×64 tile (evaluated in Table 10).

3 Experimental Results and Algorithmic Audit

The experiments below are presented in dependency order, not numerical order: $1 \to 2 \to 6 \to 4f/4g \to 3 \to 5 \to 4$.

3.1 Experiment 1: Legacy 13-Stack Benchmark and Shortcut Audit

Table 1 reports the 3-seed evaluation (`1234, 2025, 3141`) of all 13 canonical stacks and 4 un-gated (γ = None) ablations on the legacy synthetic task ($V = 24, D = 32, T = 15$, 2,000 epochs, where KEY $\in \{1..5\}$ is at $t = 0$, VAL = KEY + 5 $\in \{6..10\}$ is at $t = 1$, filler $\in \{1..23\}$ is at 2..13, and QUERY = KEY is at $t = 14$).

Table 1: Experiment 1: Full 13-stack legacy benchmark and 4 un-gated ablations (3 seeds: `1234, 2025, 3141`, mean ± std). $W_{t,0}$ and $W_{t,1}$ denote query attention weight on the key slot ($t = 0$) vs. value slot ($t = 1$). Filter labels omit the default peak gate ($\gamma = 0.5$); the last four rows use γ = None. Source: `results/exp1_legacy_13stacks.json`.

Stack	Filter	Clean (%)	KR (%)	d@0.05	d@0.5	d@1.0	nd1/8 (%)	$\sum W_m$	$\sum W_n$	Gap	$W_{t,0}$	$W_{t,1}$
`sm`	None	71.5 ± 1.4	45.1	91.7	89.1	86.8	91.4/100.0	1.000	1.000	+0.000	0.130	0.136
`sm_fft`	FFT (Non-Causal)	99.0 ± 0.1	49.8	99.1	99.2	99.2	99.1/99.8	1.000	1.000	+0.000	0.219	0.068
`sm_fir`	FIR-7	$\mathbf{100.0 \pm 0.0}$	21.9	100.0	100.0	100.0	100.0/100.0	1.000	1.000	+0.000	0.984	0.013
`pn`	None	50.6 ± 3.9	43.2	57.9	55.6	53.0	57.2/74.6	0.973	0.955	+0.001	0.074	0.227
`pn_c`	Center	58.1 ± 8.6	38.2	72.3	68.1	63.3	71.6/77.3	0.993	0.988	+0.000	0.119	0.119
`pn_tau`	τ	55.9 ± 9.7	42.7	66.7	64.9	62.0	66.1/80.1	0.848	0.811	+0.002	0.085	0.154
`pn_fft`	FFT (Non-Causal)	98.9 ± 0.3	48.7	99.0	99.0	99.0	99.0/99.4	0.995	0.995	+0.000	0.222	0.075
`pn_fft_c`	FFT+C	81.6 ± 9.3	20.9	80.8	80.9	80.6	80.8/83.2	0.952	0.952	+0.002	0.256	0.028
`pn_fir`	FIR	64.4 ± 6.8	20.5	65.5	64.9	64.7	65.3/69.1	0.512	0.000	+0.375	0.389	0.001
`pn_fir_c`	FIR+C	$\mathbf{100.0 \pm 0.0}$	22.3	100.0	100.0	100.0	100.0/100.0	0.944	0.938	+0.017	0.904	0.037
`pn_fir_tau`	**FIR+τ**	$\mathbf{100.0 \pm 0.0}$	20.6	**100.0**	**100.0**	**100.0**	**100.0/100.0**	0.570	0.375	+0.191	0.554	0.012
`qe_pn`	QE+PN	71.2 ± 1.1	45.2	91.5	87.8	83.8	90.8/100.0	0.941	0.915	+0.000	0.136	0.140
`qe_pn_fft`	QE+PN+FFT	99.6 ± 0.0	41.4	99.7	99.7	99.7	99.7/99.0	0.913	0.907	+0.003	0.243	0.074
`pn_ungated`	None (γ=None)	49.7 ± 4.1	35.5	62.9	58.1	53.7	61.8/78.1	0.999	0.992	+0.001	0.136	0.152
`pn_fir_ungated`	FIR (γ=None)	71.7 ± 19.5	21.3	71.8	71.7	71.8	71.8/72.5	0.564	0.000	+0.517	0.414	0.092
`pn_fir_c_ungated`	FIR+C (γ=None)	100.0 ± 0.0	18.7	100.0	100.0	100.0	100.0/100.0	1.000	1.000	+0.000	0.998	0.002
`pn_fir_tau_ungated`	FIR+τ (γ=None)	100.0 ± 0.0	18.7	100.0	100.0	99.9	100.0/100.0	1.000	0.717	+0.282	0.999	0.001

Audit of Exp 1 Shortcuts. Inspection of $W_{t,0}$ vs. $W_{t,1}$ in Table 1 exposes two shortcuts in the legacy task: (1) because VAL = KEY + 5 is a fixed bijection and KEY is always at $t = 0$, left-padded causal FIR acts as a boundary detector for $t = 0$ ($W_{t,0} = 0.984$ for `sm_fir`, 0.554 of 0.570 for `pn_fir_tau`), predicting VAL directly from $V_0 = W_V E(\text{KEY})$ without reading $t = 1$; and (2) because clean filler at 2..13 draws from $\{1..23\}$ (which includes competing keys $\{1..5\}$), injecting $n_d = 8$ copies of $E(\text{KEY})$ overwrites competing keys with copies of the true key, explaining why `sm` rises from 71.5% (clean) to 100.0% ($n_d = 8$).

3.2 Experiment 2: True Content-Addressable Non-Bijective Recall & Distractor Decoys

To eliminate both shortcuts, Experiment 2 places $M = 3$ randomly paired (k_m, v_m) pairs ($k_m \in \{1..8\}$, independently permuted $v_m \in \{9..16\}$) at random non-overlapping interior positions $j_m \in [2, 38]$ in $T = 48$, with filler strictly from $\{17..23\}$. True distractor decoys overwrite $n_d \in \{1, 2, 4, 8\}$ isolated filler slots with $(1 - \epsilon)E(k^*) + \epsilon E(\text{filler})$. We also evaluate `pn_fir_tau_reg`, which adds a post-warmup (epoch ≥ 1000) silence penalty $\lambda_{\text{sil}} = 0.05$ on auxiliary absent-key queries.

Table 2: Experiment 2: True Content-Addressable Non-Bijective Associative Recall ($M = 3$ pairs in $T = 48$, 3 seeds: `1234, 2025, 3141`, mean ± std). Here W_{t,j^*} and W_{t,j^*+1} measure attention on the interior target key slot vs. target value slot. Source: `results/exp2_content_addressable_decoy.json`.

Model	Clean (%)	ϵ=0.05	ϵ=0.2	ϵ=0.5	ϵ=1.0	nd1/2/4/8	W_{t,j^*}	W_{t,j^*+1}	$\sum W_m$	$\sum W_n$	Gap
`sm`	44.5 ± 0.9	47.6	44.5	43.8	46.5	45.5/45.7/44.6/46.8	0.035	0.234	1.000	1.000	$+0.000 \pm 0.000$
`pn`	35.8 ± 1.0	35.9	35.4	34.4	35.1	35.1/36.6/36.1/35.8	0.015	0.237	1.000	1.000	$+0.000 \pm 0.000$
`sm_fir`	$\mathbf{100.0 \pm 0.0}$	**95.9**	**95.9**	95.2	92.0	95.1/89.9/77.7/56.2	0.132	**0.792**	1.000	1.000	$+0.000 \pm 0.000$
`pn_fir`	88.8 ± 9.2	85.9	85.5	84.7	82.1	84.1/80.0/71.0/61.2	0.032	0.830	1.000	0.261	$+0.739 \pm 0.117$
`pn_fir_c`	96.5 ± 4.9	93.7	93.4	92.8	90.1	92.5/88.9/78.2/63.2	0.044	0.866	1.000	1.000	$+0.000 \pm 0.000$
`pn_fir_tau`	$\mathbf{100.0 \pm 0.0}$	95.5	**95.8**	**95.5**	**93.1**	**95.6/90.5/74.7/56.9**	0.063	**0.729**	0.826	0.218	$+0.608 \pm 0.135$
`pn_fir_tau_reg`	$\mathbf{100.0 \pm 0.0}$	93.8	93.1	91.8	87.4	93.1/82.6/60.6/38.8	0.034	**0.647**	0.684	**0.000**	$\mathbf{+0.684 \pm 0.030}$

Table 2 and Figures 4–5 confirm that when the bijection and boundary shortcuts are removed: (1) unfiltered `sm` (44.5%) and `pn` (35.8%) fail; (2) causal FIR on K physically binds $h_1 K(k^*)$ onto the value slot $j^* + 1$, shifting attention directly to $j^* + 1$ ($W_{t,j^*+1} = 0.792$ for `sm_fir`, 0.729 for `pn_fir_tau`, 0.647 for `pn_fir_tau_reg`); and (3) `pn_fir_tau_reg` learns $\tau = +0.115 \pm 0.021$, driving both noise and absent-key attention mass to 0.000 ± 0.000 across all seeds (gap $= +0.684 \pm 0.030$).

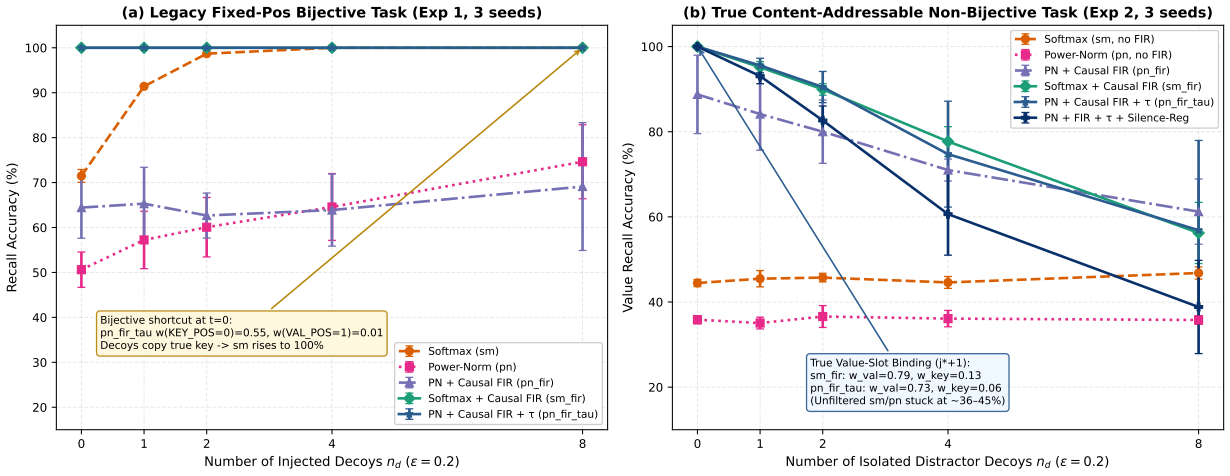


Figure 4: Multi-decoy scaling ($n_d \in \{0, 1, 2, 4, 8\}$) and attention slot routing on the legacy fixed-position task (Exp 1, left) vs. the true content-addressable non-bijective task (Exp 2, right). Source: `results/exp1_legacy_13stacks.json` and `results/exp2_content_addressable_decoy.json`.

3.3 Experiment 6: Token-Shift Ablation (How Much Memory Does Robustness Need)?

Exp 2's flooding result raises a precise follow-up: is the 7-tap low-pass doing work a 2-tap RWKV-style token shift cannot? `experiments/exp6_token_shift_ablation.py` reruns both decoy proto-

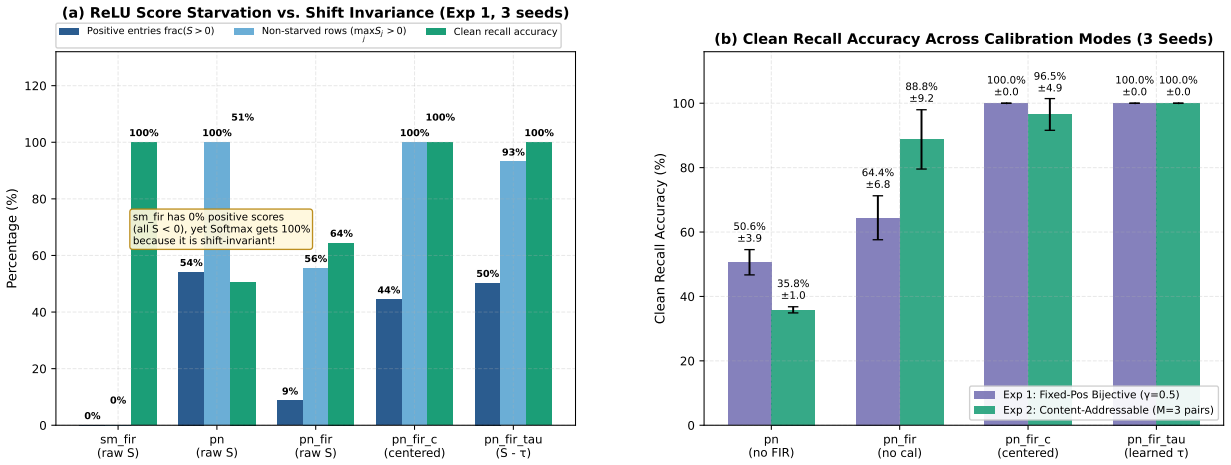


Figure 5: Pre-activation score-shift pathology in Exp 1 (left) and 3-seed clean recall accuracy across calibration modes in Exp 1 and Exp 2 (right). Source: `results/exp1_legacy_13stacks.json` and `results/exp2_content_addressable_decoy.json`.

cols with the filter on K swapped for a fixed 2-tap shift $[0.5, 0.5]$ and a learnable per-channel μ shift, at the exact FIR position (placement fixed; pre-projection placement untested). Controls are rerun in-script (`sm`/`sm_fir` reproduce Exp 1 bit-for-bit); all shift arms pass the causal probe.

Table 3: Token-shift ablation (3 seeds; `exp6`). Binding needs lag-1 memory; flooding favours shorter memory; rectified short-memory arms are seed-unstable.

Arm	Exp 1 clean	Exp 1 d@1.0	Exp 1 nd_8	Exp 2 clean	Exp 2 $\text{eps}_{0.2}$	Exp 2 nd_8	Exp 2 W_{val}
`sm`	71.5 ± 1.4	86.8 ± 1.3	100.0 ± 0.0	44.5 ± 0.9	44.5 ± 1.8	46.8 ± 1.4	0.234 ± 0.014
`sm_fir` (7-tap)	$\mathbf{100.0 \pm 0.0}$	$\mathbf{100.0 \pm 0.0}$	$\mathbf{100.0 \pm 0.0}$	$\mathbf{100.0 \pm 0.0}$	95.9 ± 0.7	56.2 ± 7.2	0.792 ± 0.011
`sm_shiftmu` (μ=0.465)	$\mathbf{100.0 \pm 0.0}$	96.9 ± 0.6	$\mathbf{100.0 \pm 0.0}$	$\mathbf{100.0 \pm 0.0}$	$\mathbf{100.0 \pm 0.0}$	$\mathbf{99.9 \pm 0.2}$	$\mathbf{0.970 \pm 0.002}$
`pn_fir_tau` (7-tap)	$\mathbf{100.0 \pm 0.0}$	99.7 ± 0.4	$\mathbf{100.0 \pm 0.0}$	$\mathbf{100.0 \pm 0.0}$	95.7 ± 2.0	56.7 ± 21.1	0.719 ± 0.021
`pn_shiftmu_tau`	76.5 ± 10.9	77.1 ± 11.1	81.8 ± 9.3	92.2 ± 11.0	92.3 ± 11.0	92.7 ± 10.4	0.798 ± 0.094

Three findings, summarized in Table 3 and Fig. 6. **(1) Binding needs only lag-1 memory:** both shift variants reach 100.0% clean recall on both protocols; learned μ pins at 0.465 ± 0.002 on Exp 2 (channels $[0.451, 0.492]$) but wanders on the legacy task ($[0.366, 0.600]$), consistent with its boundary shortcut. **(2) Flooding favours shorter memory:** at n_d=8, `sm_shiftmu` holds $99.9\% \pm 0.2\%$ vs. $56.2\% \pm 7.2\%$ for 7-tap `sm_fir` ($W_{\text{val}} = 0.970$ vs. 0.792). **(3) For rectified attention the 7-tap filter is the stable partner of a single scalar τ:** short-memory rectified arms are seed-unstable (`pn_shiftmu_tau` clean $76.5\% \pm 10.9\%$ on Exp 1 with per-seed τ–μ equilibria); the 7-tap filter's large consistent shift is reliably absorbed by one scalar.

3.4 Experiment 4f-A/4g: Width Envelope and the Tau Fixed Point

The pathology amplifies with width rather than attenuating (ladder $d \in \{16, 24, 32, 48, 64, 96, 128\}$, identical recipe, 3 seeds; `exp4f_scaleout.py` Part A and `exp4g_tau_scaling_law.py`; Fig. 7 and Fig. 8; Table 4 lists the ladder values). `sm_fir` holds 100.0% at every width (its raw shift drifts mildly, $-14 \to -27$); uncalibrated `pn_fir` raw magnitudes diverge to tens and hundreds with seed-divergent signs ($\mu_S = -20 \cdots -144$, σ_S up to $\sim$300), and `pn_fir_tau` converges fully only at $d \in \{32, 48, 64\}$ (collapsing to 52.6% at d=128; partial at 16/24, stochastic at 96/128). A linear drift-cancellation law $\tau^*(d) = a \cdot (-\mu_S) + b$ is rejected ($R^2 = 0.45$, slope ≈ 0); instead the calibrated fixed point is width-invariant ($\tau^* \in [-1.4, -0.7]$ wherever convergence happens) while only its basin

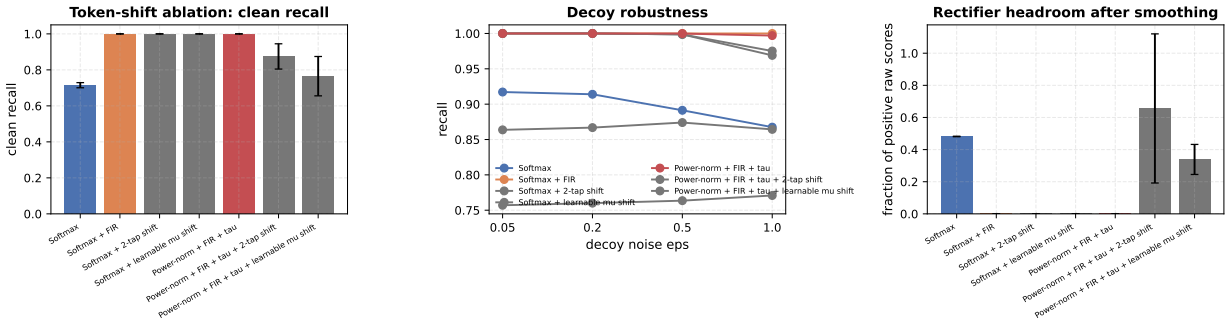


Figure 6: Token-shift ablation on the legacy protocol. Left: clean recall per variant. Middle: recall against the decoy-noise level. Right: the fraction of positive raw scores, i.e. the headroom the rectifier has left after smoothing. Bars and curves are means over $n = 3$ seeds; error bars are the standard deviation over the same seeds. Source: `results/exp6_token_shift_ablation.json` (`partA_legacy.summary`).

from $\tau_0 = 0$ collapses. Initializing the predicted $\tau_0 = -1.02$ rescues d=128 on all 3 seeds (100.0%, equilibria $-1.19/-1.16/-1.16$) — a dead-row basin problem at initialization, not a representational wall. Practical rule: start τ negative at width. (The H-gap diagnostic, $H_{\text{noise}} - H_{\text{match}}$, is ≈ 0 at every width including converged ones — uninformative under p=10 sharpening, a negative result on our own design.)

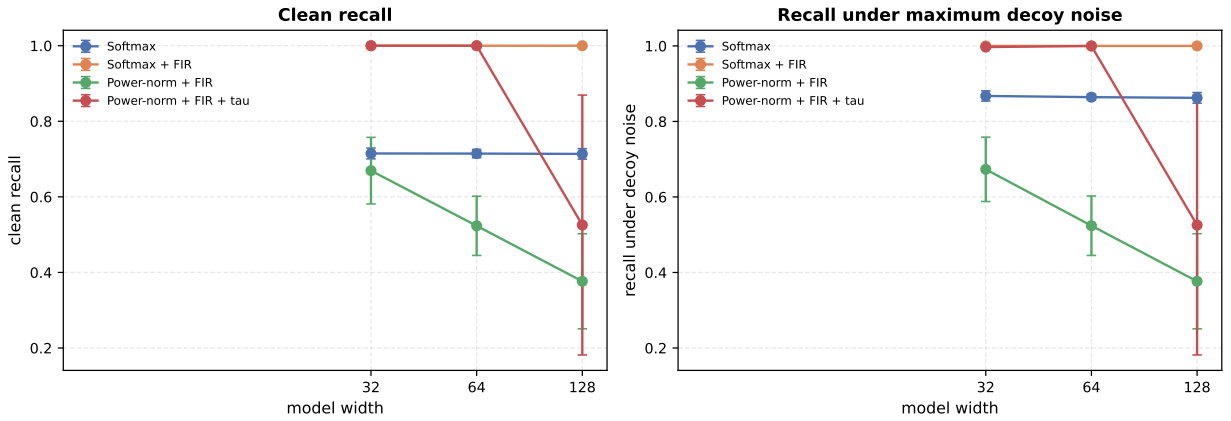


Figure 7: Width envelope of the four attention variants on the legacy task. Left: clean recall; right: recall under the strongest decoy noise. Points are means over $n = 3$ seeds; error bars are the standard deviation over the same seeds. Source: `results/exp4f_scaleout.json` (`partA.arms.<arm>.by_width.<width>`).

3.5 Experiment 3: 31-Configuration Sweep of $(\tau_0, \gamma,$ silence_mode$)$

Table 5 and Figure 9b summarize the 31-configuration sweep across threshold initialization τ_0, gate constant γ, and gating mode (`peak` vs. `mass_relu`). Initializing $\tau_0 = +0.5$ causes immediate ReLU starvation (clean = 20.6%), while $\tau_0 = 0.0$ with `peak` gating ($\gamma \in [0.1, 0.5]$) achieves 100.0% recall and strong silence (gap $= +0.572$ at $\gamma = 0.10$ and $+0.432$ at $\gamma = 0.50$ on seed 1234). In contrast, `mass_relu` (gap $\leq +0.010$) and `pn_fir_c` on positive-only training (gap $\leq +0.021$, because uncalibrated noise rows have large within-row dispersion $\max_j S_{t,j} - \mu_t \gg \gamma$) fail to suppress noise across all γ.

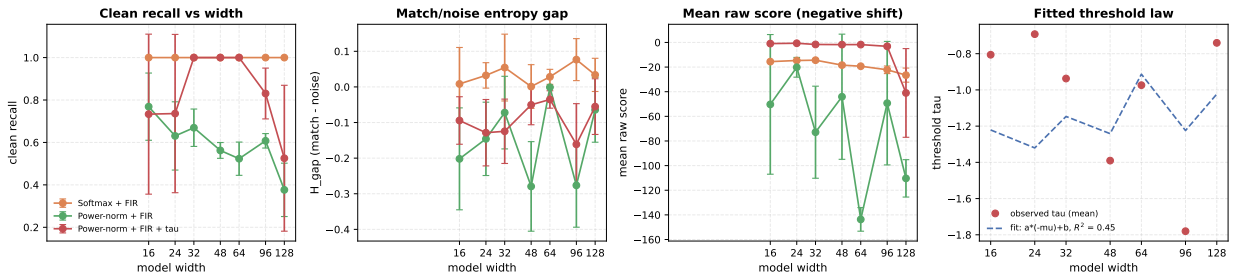


Figure 8: Behaviour of the three filtered variants across a seven-point width ladder, with the fitted threshold law. From left to right: clean recall, the match/noise entropy gap, the mean raw score, and the learned threshold against the fitted law $\tau = a \cdot (-\mu_S) + b$ ($R^2 = 0.45$, fitted on widths 32, 48, 64). Points are means over $n = 3$ seeds; error bars are the standard deviation over the same seeds. Source: `results/exp4g_tau_scaling_law.json` (`ladder`, `fit`).

Table 4: Experiment 4f-A: width envelope of the four attention variants on the legacy task (ladder $d \in \{16, 24, 32, 48, 64, 96, 128\}$, $n = 3$ seeds). μ_S is the mean raw score of the uncalibrated `pn_fir` arm, τ^* the learned threshold of `pn_fir_tau`, and the H-gap the match/noise entropy difference. The last two rows report the predicted negative initialisation $\tau_0 = -1.02$ and a $10\times$ smaller learning rate at $d = 128$. Source: `results/exp4f_scaleout.json` and `results/exp4g_tau_scaling_law.json`.

Width d	`sm_fir` clean	`pn_fir` clean	`pn_fir` raw μ_S	`pn_fir_tau` clean	`pn_fir_tau` τ^*	H-gap (tau arm)
16	100.0	76.9	−50.3	73.3	-0.81 ± 0.60	−0.09
24	100.0	63.1	−20.2	73.6	-0.69 ± 0.52	−0.13
32	100.0	66.9	−72.9	100.0	-0.94 ± 0.02	−0.13
48	100.0	56.2	−44.1	100.0	-1.39 ± 0.48	−0.05
64	100.0	52.3	−143.7	100.0	-0.97 ± 0.05	−0.04
96	100.0	60.8	−49.3	83.1	-1.78 ± 0.74	−0.16
128	100.0	37.7	−110.4	52.6	-0.74 ± 0.19	−0.06
128, $\tau_0 = -1.02$ (predicted)	—	—	—	100.0	-1.17 ± 0.02	−0.02
128, lr $= 10^{-3}$	100.0	—	—	91.0 ± 12.8	-0.60 ± 0.14	—

Table 5: Experiment 3: Representative configurations from the 31-point (τ_0, γ, mode) sweep. Source: `results/exp3_sweep_tau_gate.json`.

Stack and Mode	τ_0	γ	Clean	$\sum W_m$	$\sum W_n$	Gap	τ	3-Seed Clean	3-Seed $\sum W_n$	3-Seed Gap
`pn_fir_tau` (peak)	0.0	None	100.0%	1.000	0.752	+0.228	−0.320	100.0%	0.717	$+0.282 \pm 0.045$
`pn_fir_tau` (peak)	0.0	0.05	100.0%	0.882	0.608	+0.221	−0.453	100.0%	0.628	$+0.197 \pm 0.054$
`pn_fir_tau` (peak)	**0.0**	**0.10**	**100.0%**	**0.572**	**0.000**	**+0.572**	**−0.501**	**100.0%**	**0.336**	$\mathbf{+0.373 \pm 0.180}$
`pn_fir_tau` (peak)	0.0	0.20	100.0%	0.752	0.501	+0.211	−0.696	100.0%	0.583	$+0.129 \pm 0.070$
`pn_fir_tau` (peak)	**0.0**	**0.50**	**100.0%**	**0.436**	**0.003**	**+0.432**	**−0.940**	**100.0%**	**0.375**	$\mathbf{+0.191 \pm 0.171}$
`pn_fir_tau` (peak)	−0.5	0.50	100.0%	0.672	0.433	+0.193	−1.159	100.0%	0.484	$+0.165 \pm 0.029$
`pn_fir_tau` (peak)	−1.0	0.50	100.0%	0.724	0.611	+0.108	−1.531	100.0%	0.593	$+0.122 \pm 0.019$
`pn_fir_tau` (peak)	+0.5	0.50	20.9%	0.000	0.000	+0.000	+0.500	20.6%	0.000	$+0.000 \pm 0.000$
`pn_fir_c` (peak)	0.0	0.50	100.0%	0.947	0.946	+0.021	N/A	100.0%	0.938	$+0.017 \pm 0.004$
`pn_fir_tau` (mass_relu)	0.0	0.50	100.0%	0.847	0.851	−0.001	−0.481	100.0%	0.815	$+0.010 \pm 0.008$
`pn_fir_tau` (mass_relu)	0.0	1.00	100.0%	0.793	0.784	+0.003	−0.613	100.0%	0.776	$+0.008 \pm 0.006$

3.6 Experiment 5: Context Override vs. Residual Stream Preservation Against `Sparsemax`, `Entmax`, and `Softmax Sinks` Under Pure Cross-Entropy Loss

To test whether sub-simplex selective silence yields a concrete end-to-end task advantage over simplex normalizers (`sm_fir`, `Sparsemax` [9], and α-`Entmax` [10] [11]), row-mean-centered power-norm (`pn_fir_c`), and virtual-sink Softmax (`softmax1_fir`, `softmax_sink_fir`) **under 100% pure task Cross-Entropy loss with zero auxiliary attention regularization**, Experiment 5 (`exp5_abstention_and_entmax.py`) evaluates a topic-coherent document residual stream task ($T = 48$, $M = 3$ pairs $(k_m, v_m) \in \{0..7\}^2$, filler tokens carrying topic values from v_{sub}, and

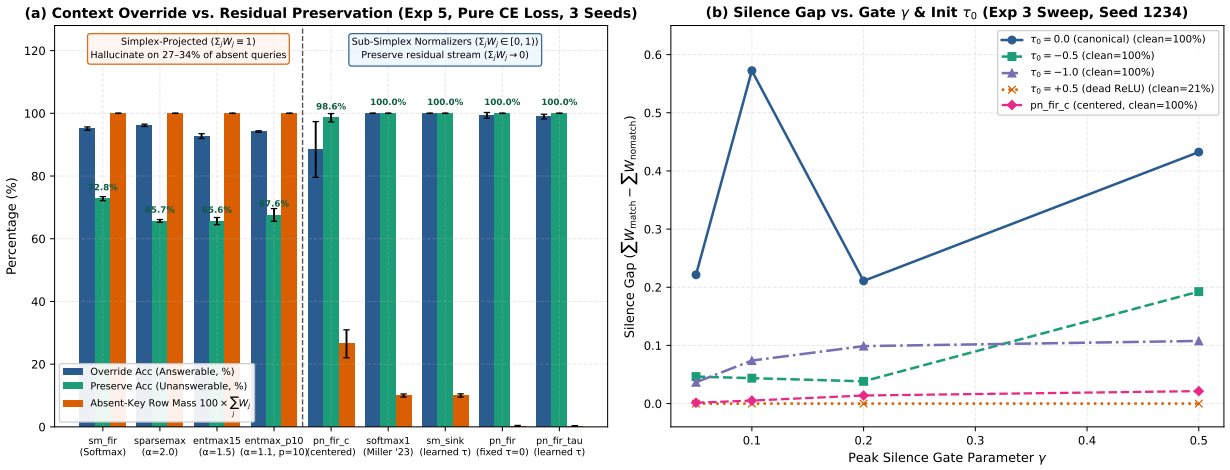


Figure 9: Context override vs. residual stream preservation across simplex and sub-simplex normalizers under pure Cross-Entropy loss (Exp 5, left), and silence gap vs. gate γ and initialization τ_0 (Exp 3 sweep, right). Source: `results/exp5_abstention_and_entmax.json` and `results/exp3_sweep_tau_gate.json`.

query residual update $y = V(v_{\text{default}}) + 2.5 \sum_{j<47} W_{\text{query},j} V_j$ where $v_{\text{default}} \notin v_{\text{sub}}$). On 50% of queries ($k^*$ present), the head must override v_{default} with v^*; on 50% of queries (k^* absent), the head must stay silent ($\sum_j W_{\text{query},j} \to 0$) to preserve v_{default}.

Table 6: Experiment 5: Context Override vs. Residual Preservation under pure Cross-Entropy loss (3 seeds: `1234, 2025, 3141`, mean ± std). All 9 models use the 7-tap causal FIR filter on K. Source: `results/exp5_abstention_and_entmax.json`.

Family	Model	Overall (%)	Overr. (%)	Pres. (%)	Hall. (%)	$\sum W_a$	$\sum W_u$	Gap	W_{t,j^*+1}
Simplex	`sm_fir`	84.0 ± 0.6	95.1 ± 0.5	72.8 ± 0.7	$\mathbf{27.2 \pm 0.7}$	1.000	1.000	+0.000	0.538
Simplex	`sparsemax_fir` ($\alpha = 2.0$)	80.9 ± 0.2	96.2 ± 0.4	65.7 ± 0.4	$\mathbf{34.3 \pm 0.4}$	1.000	1.000	+0.000	0.604
Simplex	`entmax15_fir` ($\alpha = 1.5$)	79.2 ± 0.8	92.7 ± 0.8	65.6 ± 1.1	$\mathbf{34.4 \pm 1.1}$	1.000	1.000	+0.000	0.593
Simplex	`entmax_p10_fir` ($\alpha = 1.1$)	80.9 ± 0.9	94.2 ± 0.2	67.6 ± 2.0	$\mathbf{32.4 \pm 2.0}$	1.000	1.000	+0.000	0.560
Centered	`pn_fir_c`	93.5 ± 5.1	88.5 ± 8.9	98.6 ± 1.3	1.4 ± 1.3	0.716	0.265	+0.451	0.577
Virtual Sink	`softmax1_fir` ($\tau = 0$)	$\mathbf{100.0 \pm 0.0}$	$\mathbf{100.0 \pm 0.0}$	$\mathbf{100.0 \pm 0.0}$	$\mathbf{0.0 \pm 0.0}$	0.863	0.100	+0.764	0.776
Virtual Sink	`softmax_sink_fir` (τ)	$\mathbf{100.0 \pm 0.0}$	$\mathbf{100.0 \pm 0.0}$	$\mathbf{100.0 \pm 0.0}$	$\mathbf{0.0 \pm 0.0}$	0.864	0.100	+0.764	0.775
Sub-Simplex	`pn_fir` ($\tau = 0, p = 10$)	99.7 ± 0.5	99.4 ± 0.9	$\mathbf{100.0 \pm 0.0}$	$\mathbf{0.0 \pm 0.0}$	0.747	**0.002**	+0.745	0.724
Sub-Simplex	`pn_fir_tau` ($\tau, p = 10$)	$\mathbf{99.5 \pm 0.4}$	$\mathbf{99.0 \pm 0.7}$	$\mathbf{100.0 \pm 0.0}$	$\mathbf{0.0 \pm 0.0}$	0.744	**0.002**	**+0.742**	0.716

Table 6 and Figure 9a establish four key findings: (1) all four simplex normalizers fail residual preservation (27.2%–34.4% hallucination rate), with `sparsemax_fir` and `entmax` performing even worse than `sm_fir` because sparse simplex projection concentrates weight onto a single mismatching topic token; (2) under pure Cross-Entropy loss with 50% unanswerable queries, `pn_fir_tau` places unanswerable raw score peaks below $\tau = +0.011 \pm 0.020$ ($\max_j S^{\text{raw}}_{\text{unans}} = -0.175 \pm 0.070 < \tau$, while $\max_j S^{\text{raw}}_{\text{ans}} = +1.548 \pm 0.281 \gg \tau$), achieving $100.0\% \pm 0.0\%$ residual preservation (0.0% hallucination rate) and $\sum W_{\text{unans}} = 0.002 \pm 0.002$ (note that because 50% of training examples are unanswerable, W_Q, W_K naturally center unanswerable scores below zero, so fixed $\tau = 0$ `pn_fir` performs equally well here at $99.7\% \pm 0.5\%$, whereas learnable τ is essential when training is dominated by positive retrieval as in Exp 1–2 or multi-layer LM in Exp 4); (3) `pn_fir_c` shrinks $\|Q_{\text{query}}\|_2$ on unanswerable rows to reach $\sum W_{\text{unans}} = 0.265 \pm 0.045$ (gap $= +0.451 \pm 0.032$), but still leaks $\approx 124\times$ more residual noise than `pn_fir_tau` while degrading answerable override accuracy ($88.5\% \pm 8.9\%$ vs. $99.0\% \pm 0.7\%$); and (4) comparing `pn_fir_tau` against virtual-sink Soft-

max (`softmax1_fir`, `softmax_sink_fir`) reveals a clean trade-off: virtual-sink Softmax achieves slightly higher discrete answerable override accuracy ($100.0\% \pm 0.0\%$ vs. $99.0\% \pm 0.7\%$) and higher active mass ($\sum W_{\text{ans}} = 0.863$–$0.864$ vs. 0.744), whereas `pn_fir_tau` achieves $50\times$ lower continuous residual-stream contamination on unanswerable queries ($\sum W_{\text{unans}} = 0.002 \pm 0.002$ vs. 0.100 ± 0.005).

3.7 Experiment 4: 6-Layer Causal Language Modeling on WikiText-2 and Fused Triton Benchmarks

We evaluate a 6-layer Pre-LN Transformer on word-level WikiText-2 ($V = 23,168, D = 64, H = 4, d_{\text{ff}} = 256, N = 256, B = 64$, tied embeddings: $1,482,752$ (tok_emb) + $16,384$ (pos_emb) + $6 \times 49,728$ (298,368) + 128 (ln_f) = $1,797,632$ base parameters; +6 scalar τ_ℓ parameters in `pn_fir_tau` for $1,797,638$ total). In IEEE-754 `FP32` parity checks (`allow_tf32=False`, reference norms $\|y\|_\infty = 0.5441$, $\|dx\|_\infty = 2.1698$, $|d\tau| = 7.0289$), both `PnFirTauTritonMHA` (out_max_diff $= 2.38 \times 10^{-7}$ [rel 4.38×10^{-7}], dx_max_diff $= 1.16 \times 10^{-6}$ [rel 5.36×10^{-7}], dtau_diff $= 3.91 \times 10^{-5}$ [rel 5.56×10^{-6}]) and `PnFirTauFlashMHA` (out_max_diff $= 2.38 \times 10^{-7}$, dx_max_diff $= 1.32 \times 10^{-6}$ [rel 6.08×10^{-7}], dtau_diff $= 4.01 \times 10^{-5}$ [rel 5.70×10^{-6}]) closely match eager PyTorch. Table 7, Table 9, Table 10, and Figure 18 present the full benchmark suite.

Table 7: Experiment 4: WikiText-2 6-layer Language Model benchmarks ($B = 64, N = 256$, RTX 3060). Suites 1–3 and suite 2s are defined in the text; suite 3 is precision-controlled across 3 seeds; suite 2s (D=256) is a single-seed pilot; step times are omitted for suite 3 and suite 2s (see notes below). Source: `results/exp4_wikitext2_lm.json`.

Suite	Variant	Precision	ms/step	Rel. Time	VRAM (GB)
suite 1: Speed/VRAM	`softmax`	`BF16+TF32`	97.70	1.000×	5.16
suite 1: Speed/VRAM	`sm_fir` (FIR)	`BF16+TF32`	99.21	1.016×	5.17
suite 1: Speed/VRAM	`pn_fir_tau` (Eager)	`BF16+TF32`	164.44	1.683×	6.11
suite 1: Speed/VRAM	`pn_fir_tau_triton`	`BF16+TF32`	**75.74**	**0.775×**	**4.97**
suite 1: Speed/VRAM	`pn_fir_tau_flash`	`BF16+TF32`	**66.80**	**0.684×**	**4.52**
suite 1: Speed/VRAM	`softmax`	`FP32+TF32`	140.92	1.000×	6.92
suite 1: Speed/VRAM	`sm_fir` (FIR)	`FP32+TF32`	143.76	1.020×	6.94
suite 1: Speed/VRAM	`pn_fir_tau` (Eager)	`FP32+TF32`	244.17	1.733×	8.82
suite 1: Speed/VRAM	`pn_fir_tau_triton`	`FP32+TF32`	**120.43**	**0.855×**	**6.92**
suite 1: Speed/VRAM	`pn_fir_tau_flash`	`FP32+TF32`	**105.84**	**0.751×**	**6.10**

Suite	Variant	Precision	Params	Train	Val	Test	Leak
suite 2	`softmax`	`BF16+TF32`	1,797,632	654.23	547.33	515.55	5.16
suite 2	`sm_fir`	`BF16+TF32`	1,797,632	652.97	547.29	515.18	5.17
suite 2	`pn_fir_tau` (Eager)	`BF16+TF32`	1,797,638	653.17	546.75	515.56	6.11
suite 2	`pn_fir_tau_triton`	`BF16+TF32`	1,797,638	652.00	545.78	515.40	**4.97**
suite 2	`pn_fir_tau_flash`	`BF16+TF32`	1,797,638	653.54	546.78	516.11	**4.52**
suite 2s	`softmax`	`BF16+TF32`	10,795,008	441.16	411.03	379.83	—‡
suite 2s	`sm_fir`	`BF16+TF32`	10,795,008	416.71	390.09	359.48	—‡
suite 2s	`pn_fir_tau` (+6 τ)	`BF16+TF32`	10,795,014	428.52	409.33	377.36	—‡
suite 3	`softmax`	`BF16+TF32`	1,797,632	77.93 ± 0.41	124.37 ± 0.26	117.13 ± 0.04	—
suite 3	`pn_fir_tau`	`BF16+TF32`	**1,797,638**	78.30 ± 0.45	$\mathbf{122.15 \pm 0.72}$	$\mathbf{114.80 \pm 0.49}$	—

Precision-controlled 50k, 3 seeds. Both suite 3 runs use `BF16+TF32` across 3 seeds per variant (softmax: one seed-42 run plus seeds 43–44; pn: one seed-42 run plus seeds 43–44; `results/exp4d_softmax_50k_bf16.json`, `results/exp4e_multiseed_50k.json`), giving a matched 50k gap of -2.22 ± 0.46 val / -2.33 ± 0.49 test PPL for `pn_fir_tau`, holding on all 3 seeds (per-seed val gaps $-2.26/-2.76/-1.64$).

suite 2s notes. †Effective wall-clock per step including eval amortization (725/1121/789 ms for softmax/sm_fir/pn_fir_tau). A pure-step bench was attempted and abandoned: under WSL GPU-PV paging the FP32 suite crawls and eager `pn_fir_tau` OOMs the bench harness on backward. ‡Peak VRAM under WSL paging is non-comparable (measured 14.1/14.2/18.2 GB on a 12 GB card) and omitted; a native-Linux measurement is left for future work.

Table 8: Quantization Hygiene: INT8 Activation Noise, Attention-Sink Mass, and Residual Outliers (Exp 4h). Per-token symmetric INT8 quantize-dequantize on every `nn.Linear` input (activation-only emulation, weights kept in fp), evaluated on the trained checkpoints. Sink mass is the mean attention mass on position 0 over the six layers. Source: `results/exp4h_int8_quant.json`.

Model	FP test PPL	INT8-sim PPL	Degrad.	Sink mass pos-0	Resid. absmax	Resid. kurtosis
D=64 `softmax` 50k	117.15	117.27	+0.10%	0.0177	5.03	4.17
D=64 `pn_fir_tau` 50k	114.25	114.35	+0.09%	0.0031	4.08	4.22
D=256 `softmax` 2.4k	379.81	380.96	+0.30%	0.0122	207.3	85.6
D=256 `sm_fir` 2.4k	359.47	360.75	+0.36%	0.0191	219.7	119.7
D=256 `pn_fir_tau` 2.4k	377.37	378.88	+0.40%	0.0013	176.5	78.5

Quantization hygiene (Exp 4h). Per-token symmetric INT8 quantize-dequantize on every `nn.Linear` input (Table 8) costs $< 0.5\%$ test PPL for every variant (D=64: 117.27 vs. 117.15 softmax, 114.35 vs. 114.25 pn; D=256: +0.30/+0.36/+0.40%) — no Softpick-scale low-bit win replicates at 2–11M parameters, and none is claimed. No massive sinks appear anywhere (max pos-0 mass 1.9%). The rectified hygiene advantage is real but conditional: `pn_fir_tau` carries 6–15× less pos-0 mass (0.0031 vs. 0.0177 at D=64; 0.0013 vs. 0.0122–0.0191 at D=256, decaying to 0.0001 with depth) and 15–44% lower kurtosis across roles under D=256 scale-up stress, closing after converged training at D=64.

Table 9: Isolated 6-Layer MHA Stack and 1-Layer Kernel (`kern1`) Scaling ($D = 64, H = 4, d_k = 16$, uniform $B_M = B_N = 64$, Forward + Backward in `BF16+TF32`, 200 steps after 25 warmup, RTX 3060). Source: `results/exp4b_mha6_instrumentation.json`.

Seq, Batch	`softmax`	`sm_fir`	`pn` Eager	`pn_triton`	`pn_flash`	
$N = 256, B = 64$	34.26 ± 0.69	35.23 ± 0.80	101.02 ± 1.29	$\mathbf{21.74 \pm 4.97}$	$\mathbf{17.66 \pm 2.28}$	$[1.19 \pm 0.28]$
$N = 512, B = 64$	125.41 ± 2.34	127.52 ± 2.04	390.64 ± 6.12	$\mathbf{37.76 \pm 0.94}$	$\mathbf{20.60 \pm 2.73}$	$[2.05 \pm 0.14]$
$N = 1024, B = 64$	OOM	OOM	OOM	$\mathbf{132.23 \pm 1.01}$	$\mathbf{46.77 \pm 0.54}$	$[6.17 \pm 0.20]$
$N = 1024, B = 16$	126.35 ± 0.67	127.76 ± 0.34	393.42 ± 0.43	$\mathbf{34.33 \pm 0.52}$	$\mathbf{18.13 \pm 2.44}$	$[1.98 \pm 0.17]$

(a) step time (ms/step, fwd+bwd); (b) peak VRAM (MB). Per-row speedups (4.65×–21.70×) are reported in §3.5.

Seq, Batch	`softmax`	`sm_fir`	`pn` Eager (MB)	`pn_triton` (MB)	`pn_flash` (MB)
$N = 256, B = 64$	815.2	827.5	1886.9	**503.3**	**109.3**
$N = 512, B = 64$	3086.3	3110.6	7341.4	**1821.8**	**201.8**
$N = 1024, B = 64$	OOM (> 12 GB)	OOM (> 12 GB)	OOM (> 12 GB)	**6954.8**	**386.8**
$N = 1024, B = 16$	3028.8	3040.9	7268.3	**1751.3**	**109.3**

Table 10: Silent-Tile Skipping Profile ($N = 1024, B = 16$, $B_M = B_N = 64$, Forward + Backward, 200 steps). While the isolated 1-layer kernel (`kern1`) speeds up by 1.73×–1.76× when whole 64×64 tiles are silent ($\tau \geq +1.0$), scattered row silence ($\tau = +0.5, 69.1\%$ silent rows) leaves at least one positive entry in almost every 64×64 tile. The 6-layer module is launch-serialised (≈372 CUDA kernel launches per step, invariant across τ and $B \in \{4, 16\}$; ≈52 µs per launch under WSL2 GPU-PV), so isolated-kernel savings do not translate into stack wall time; the `mha6` column here uses the original sequential protocol and its end-to-end difference is identified as a block-order artifact by the drift-controlled re-measurement (`experiments/exp4b_mha6_instrumentation.py`).

Threshold τ	Silent rows	Mean row mass	1L kernel (ms)	1L speedup	6L module (ms)	6L speedup
−0.5	0.0%	0.655	1.99 ± 0.14	1.00×	19.25 ± 3.35	1.00×
0.0	0.3%	0.469	1.94 ± 0.15	1.02×	18.77 ± 2.90	1.03×
+0.5	69.1%	0.048	1.91 ± 0.12	1.04×	18.24 ± 2.60	1.06×
+1.0	99.9%	0.0001	$\mathbf{1.14 \pm 0.21}$	**1.73**×	18.32 ± 3.47	1.05×
+1.5	100.0%	0.0000	$\mathbf{1.13 \pm 0.20}$	**1.75**×	18.07 ± 3.11	1.07×
+2.5	100.0%	0.0000	$\mathbf{1.13 \pm 0.18}$	**1.76**×	17.65 ± 2.32	1.09×

Drift-controlled re-measurement of the 6-layer stack. To separate the kernel-level saving from the stack-level wall, we re-instrumented the exact configuration of Table 10 (N=1024, B=16, `BF16` autocast, matmul `TF32` off, seed 42) with τ blocks interleaved in rotating order (three ($\tau=-0.5, \tau=+2.5$) pairs, three low-start and three high-start; 25 warmup + 120 timed steps per block), CUDA-event spans around the forward residual loop and the backward pass, per-layer per-tile skip diagnostics, `torch.profiler` launch counts, and a $B \in \{16, 4\}$ workload-scaling discriminator. Table 11 reports the outcome: the stack wall is τ-flat (-0.43 ± 0.43 ms, $|z| \approx 1.0$, $n = 6$ block pairs) even though the isolated kernel is not ($2.167 \pm 0.027 \rightarrow 1.464 \pm 0.032$ ms, 1.48×, i.e. 0.703 ± 0.042 ms per layer, 4.2 ms over six layers), the number of CUDA kernel launches per step is invariant across τ and B (372.7 vs. 372.3), and 100.0% of the 64×64 tiles are fully silent in every layer at $\tau = +2.5$ against the 46.9% causal floor at $\tau = -0.5$. The launch-bound diagnosis is then confirmed by intervention: capturing the entire 6-layer forward+backward in a CUDA graph (Table 12) amortises dispatch to a single replay and surfaces the saving in full ($12.10 \pm 0.03 \rightarrow 4.91 \pm 0.02$ ms, $\Delta = -7.20 \pm 0.02$ ms), while the eager walls measured in the same process stay τ-flat (+2.04 ms, within drift). The design lever at this stack scale is therefore fewer launches per step (graph capture or kernel fusion), not more skipping.

Table 11: Drift-Controlled Re-Measurement of the 6-Layer Stack (Exp 4b). The τ blocks are interleaved in rotating order (three ($\tau=-0.5, \tau=+2.5$) pairs, three low-start and three high-start; 25 warmup + 120 timed steps per block) at N=1024, B=16, `BF16` autocast, matmul `TF32` off, seed 42, so that the wall time is measured against within-session drift instead of against the block order. Wall times, spans and kernel times are in milliseconds; the last row is the B 16 → 4 workload-scaling comparison (standard errors over the two blocks: $\tau=-0.5$: 19.50 ± 0.20 vs. 18.88 ± 0.16 ms; $\tau=+2.5$: 18.49 ± 0.39 vs. 19.76 ± 0.36 ms). Source: `results/exp4b_mha6_instrumentation.json`.

Quantity	$\tau = -0.5$ (active)	$\tau = +2.5$ (silent)	Δ
Stack wall per step (6 blocks)	20.30 ± 0.30	19.87 ± 0.33	-0.43 ± 0.43 ($\lvert z \rvert \approx 1.0$)
Forward residual-loop span	8.23	8.37	+0.15 (τ-invariant)
Backward span	11.61	10.96	−0.65 (vs. 4.2 predicted)
Isolated kernel wall (`kern1`)	2.167 ± 0.027	1.464 ± 0.032	-0.703 ± 0.042 (1.48×)
Fully-silent 64×64 tiles/layer	46.9%	100.0%	+53.1 p.p.
CUDA kernel launches per step	372.7	372.3	invariant ($\approx 52\,\mu$s/launch)
Stack wall, 4× less GPU work	19.50 vs. 18.88	18.49 vs. 19.76	3.08× / 3.23× SEM

Table 12: Launch Amortisation Isolated by CUDA-Graph Replay (Exp 4c). The whole 6-layer forward+backward is captured in a CUDA graph, so per-step launch dispatch is replaced by a single replay; the eager walls are measured in the same process. The replay-determinism check gives a maximum relative gradient deviation of 5.5×10^{-3} at $\tau = -0.5$ and exactly 0 at $\tau = +2.5$. Source: `results/exp4c_cuda_graphs.json`.

Quantity (rotated interleave)	$\tau = -0.5$ (fully active)	$\tau = +2.5$ (all tiles silent)	Δ (+2.5 vs. −0.5)
Eager stack wall (launch-bound)	21.46 ± 0.47 ms	23.50 ± 0.68 ms	+2.04 ms (within drift)
CUDA-graph replay wall (fwd + bwd)	12.10 ± 0.03 ms	4.91 ± 0.01 ms	-7.20 ± 0.02 ms
Speed-up vs. eager at the same τ	1.77×	4.79×	—

Supporting figures for Experiment 4. Figures 10 and 11 show the 50k-step training curves and the per-seed perplexity of the matched precision-controlled protocol. Figure 12 reports the per-layer silence diagnostics behind Table 10, and Figure 13 illustrates the silent-tile skipping mechanism. Figure 14 compares the saving predicted from the single-kernel measurement against the saving realized by the 6-layer stack, Figure 15 isolates launch amortisation by CUDA-graph replay, and

Figure 16 quantifies the measurement-order drift. Figure 17 reports the INT8 outlier and attention-sink diagnostics of Experiment 4h.

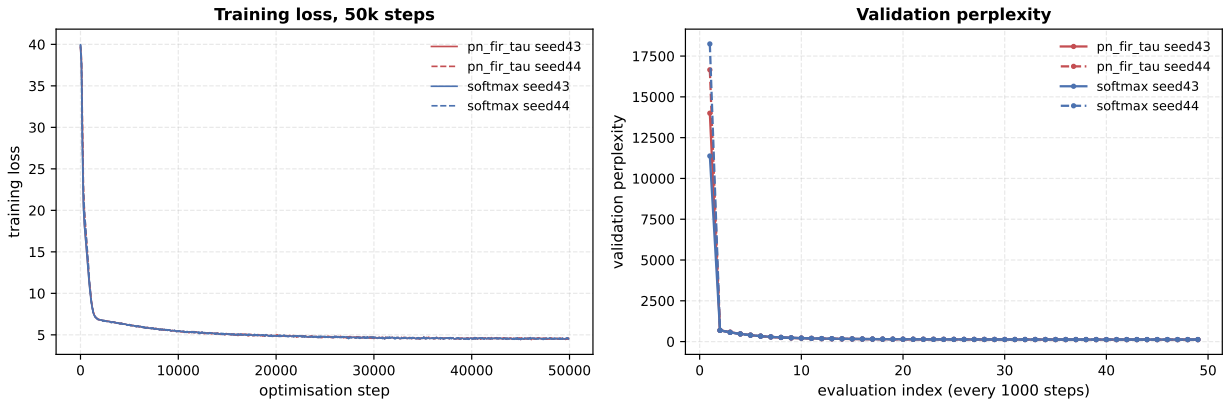


Figure 10: Training history of the 50k-step runs. Left: training loss against optimisation step. Right: validation perplexity at the periodic evaluations. Curves are $n = 2$ seeds per variant (seeds 43 and 44); no seed-level error band is shown, because two seeds do not support one. Source: `results/exp4e_multiseed_50k.json` (`runs.<variant>/<seed>.history`).

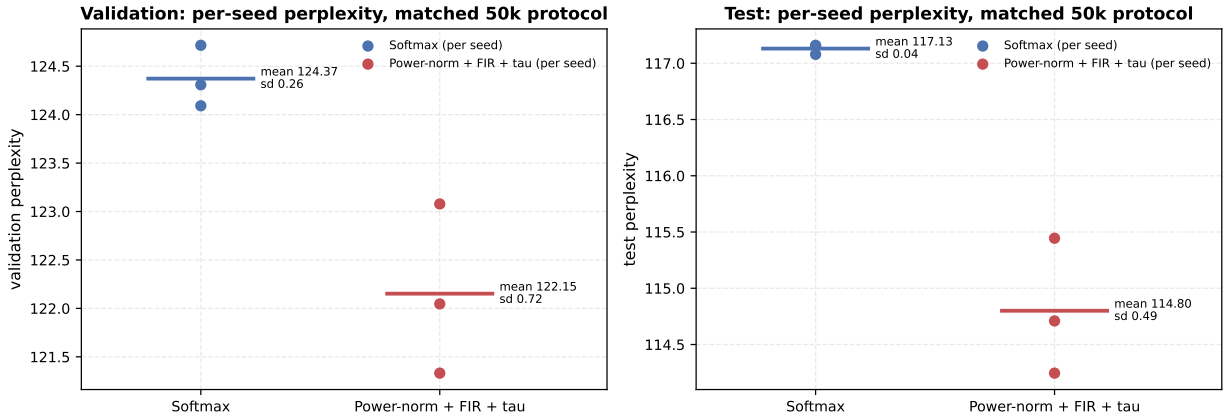


Figure 11: Per-seed perplexity under the matched 50k protocol. Each dot is one seed ($n = 3$ per variant); the bar is the mean over those seeds and the annotation gives the mean and the standard deviation over seeds. The seed-42 point of the rectified variant is the reference run of the matched-precision record; the two remaining seeds come from the multi-seed run. Source: `results/exp4e_multiseed_50k.json` and `results/exp4d_softmax_50k_bf16.json`.

4 Limitations and Negative Results

Six limitations bound the claims above.

1. **Scale:** all multi-seed language-modeling results are at D=64 (1.8M parameters, $L = 6$, seq=256); a single-seed scale-out at D=256/N=512 (10.8M, suite 2s: `sm_fir` 359.5 test, `pn_fir_tau` 377.4, `softmax` 379.8) shows no collapse and preserves the ranking, with a replicated tau profile (layer 0 at -0.50). An auxiliary GPT-2 124M retrofit probe (Softmax→`pn_fir_tau` weight transplant, 20k fine-tune steps) reached Test PPL 47.05 vs. 21.43 for a Softmax retrofit of the same budget; its raw artifacts are not distributed with this work. We attribute the gap to co-adaptation: pre-trained W_Q, W_K are optimized for the

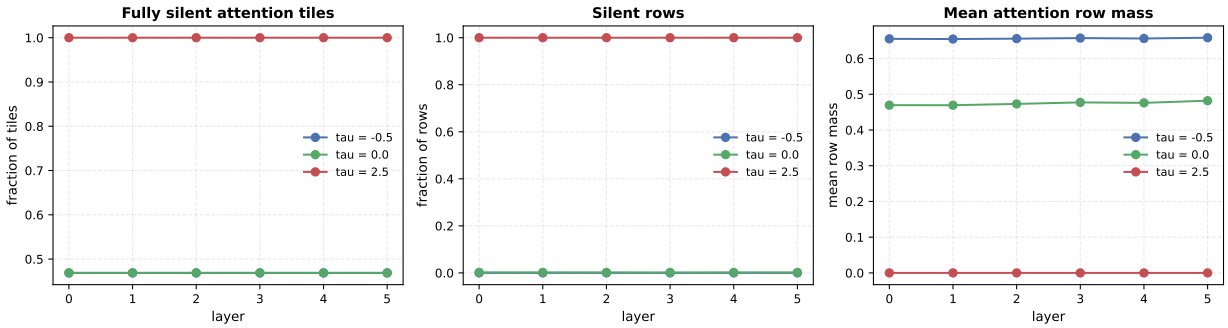


Figure 12: Per-layer silence diagnostics of the 6-layer stack for three thresholds. Left: fraction of fully silent attention tiles. Middle: fraction of silent rows. Right: mean attention row mass. Single instrumented run per threshold (one run per τ), so no seed spread is shown. Source: `results/exp4b_mha6_instrumentation.json` (`diagnostics_per_tau`).

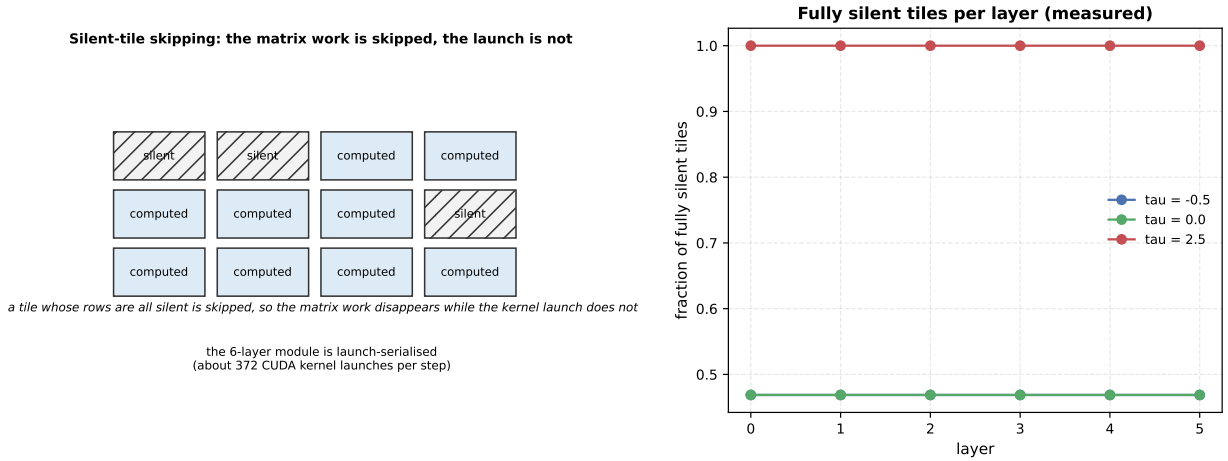


Figure 13: Schematic of silent-tile skipping. Left: a tile grid in which tiles whose rows are all silent are skipped, so the matrix work disappears while the kernel launch does not; the 6-layer module is launch-serialised with about 372 CUDA kernel launches per step. Right: the measured fraction of fully silent tiles per layer and threshold. Single instrumented run per threshold. Source: `results/exp4b_mha6_instrumentation.json` (`diagnostics_per_tau`) and `src/triton_pnfirtau.py`.

translation-invariant geometry of the Softmax simplex, whereas the FIR-induced score shift changes that geometry; a 20k fine-tune cannot re-learn the projection matrices, so the gap is expected. Continuous pre-training or low-rank/evolutionary adaptation is the appropriate protocol at this scale; no GPT-2-class result is claimed. At 1.8M parameters attention sinks are present but modest (Exp 4h: max position-0 mass 1.9%), so sink-suppression benefits are bounded at this scale. RAG-style behavior is evaluated on a controlled synthetic task, not on a production retrieval corpus.

2. **Matched 50k with variance, small scale:** Table 7 (suite 3) is precision-controlled across 3 seeds per variant, with a -2.22 ± 0.46 val / -2.33 ± 0.49 test PPL gap for `pn_fir_tau`, holding on all 3 seeds; pn shows larger seed spread than softmax (val std 0.72 vs. 0.26), and all results are at D=64 with no larger-model replication. All quoted throughput numbers are single-seed.

3. **Tile-granularity and launch serialisation:** silent-tile skipping accelerates the isolated

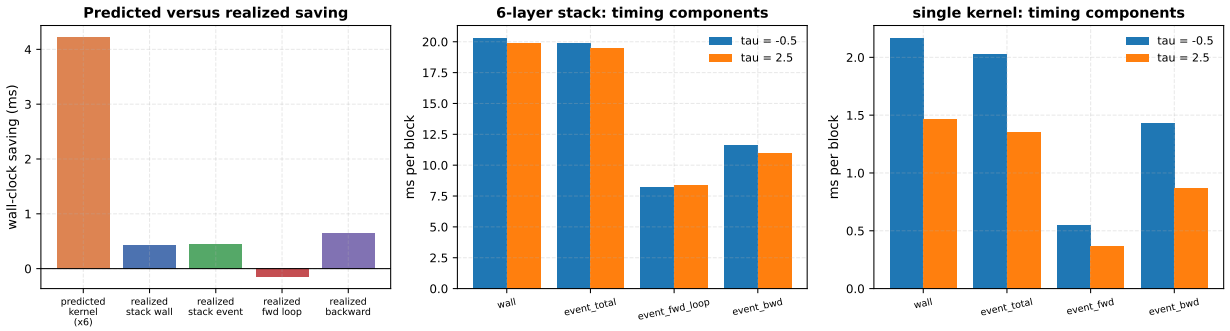


Figure 14: Predicted versus realized wall-clock saving. Left: the saving predicted from the single-kernel measurement against the saving actually realized by the 6-layer stack and its forward/backward split. Centre and right: timing components of the stack and of the single kernel. Single instrumented run per threshold; the error bars of the underlying block timings are reported in the text, not drawn here. Source: `results/exp4b_mha6_instrumentation.json` (`attribution`, `stack_timing`, `kern1_timing`).

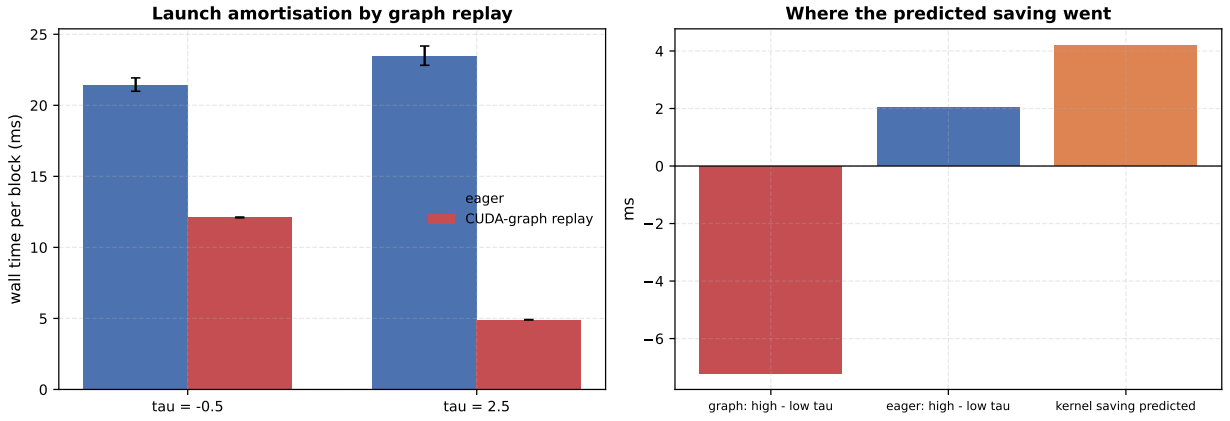


Figure 15: Launch amortisation isolated by CUDA-graph replay. Left: wall time per block of the 6-layer stack under eager execution and under graph replay, for two thresholds; error bars are the standard error over the measured blocks of a single run. Right: the high-minus-low threshold difference under graph replay and under eager execution, against the saving predicted from the kernel measurement. Source: `results/exp4c_cuda_graphs.json` (`combined`, `attribution`).

kernel (1.48× re-measured, 1.76× original) but is launch-serialised in the 6-layer stack (≈372 CUDA launches per step, invariant across τ and $B \in \{4, 16\}$; wall unchanged at 4× less GPU work under WSL2 GPU-PV). The end-to-end stack saving attributed to tile skipping (10, `mha6` column) was reproduced and falsified as a block-order artifact by the drift-controlled re-measurement (`exp4b`). A CUDA-graph replay confirms the mechanism by intervention: amortising launches surfaces the saving fully ($12.10 \pm 0.03 \rightarrow 4.91 \pm 0.02$ ms, $\Delta = -7.20 \pm 0.02$ ms, `exp4c`).

4. **Decoy flooding is memory-length-specific:** exact copies of the queried key at high multiplicity ($n_d \geq 4$; 56.9% at n_d=8, ϵ=0.2 under the 7-tap FIR) do not defeat short-memory linear filtering: a 2-tap shift holds 99.9%±0.2% (`sm_shiftmu`) and 92.7%±10.4% (`pn_shiftmu_tau`) at n_d=8 (Exp 6), so the failure is specific to the 7-tap memory length, at the cost of seed-dependent τ–μ equilibria in the rectified branch.

5. τ **calibration regime:** under positive-only training, gradients on τ come from active rows

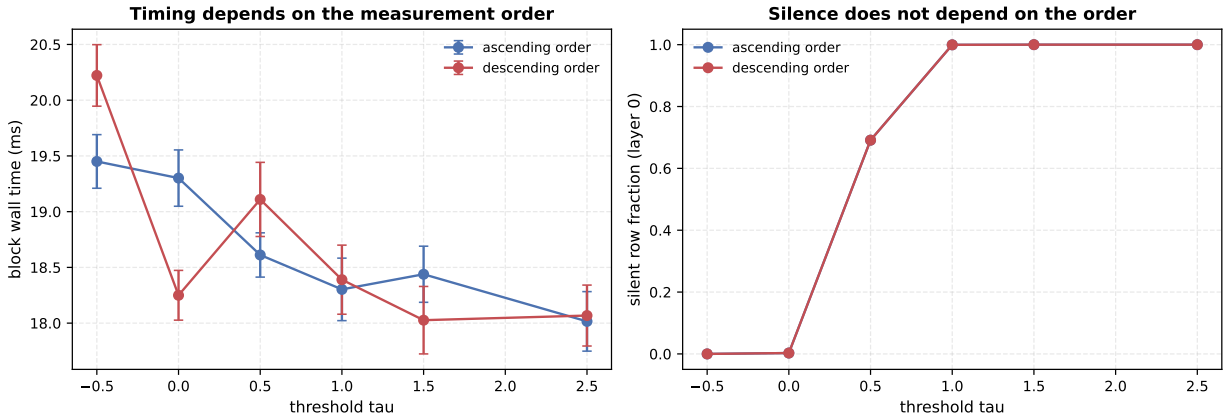


Figure 16: Sensitivity of the block timing to the order in which the thresholds are measured. Left: block wall time against threshold, one line per measurement order; error bars are the standard error over the blocks of a single run. Right: the silent-row fraction of the first layer, which does not depend on the order. Source: `results/exp4b_sequential_repro.json` (`runs`).

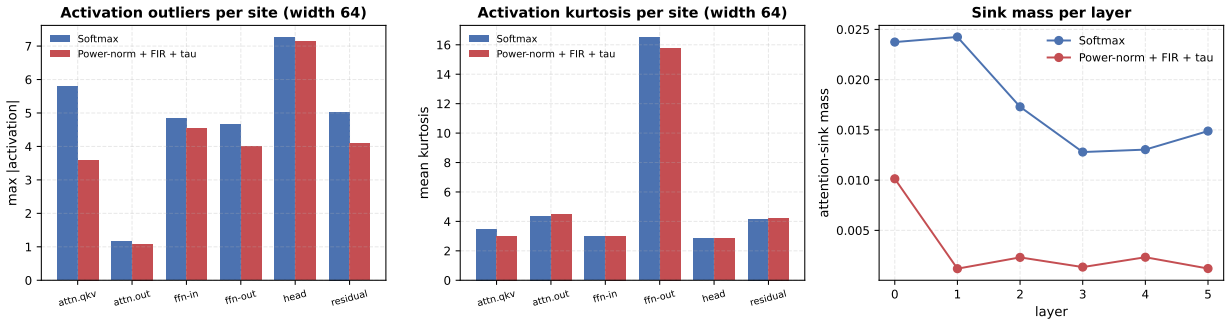


Figure 17: Activation outliers and attention-sink mass of the width-64 runs. Left: maximum absolute activation per site. Middle: mean kurtosis per site. Right: sink mass per layer. One trained checkpoint per variant (no seed spread available). Source: `results/exp4h_int8_quant.json` (`runs.<run_id>.outliers`, `runs.<run_id>.sink`).

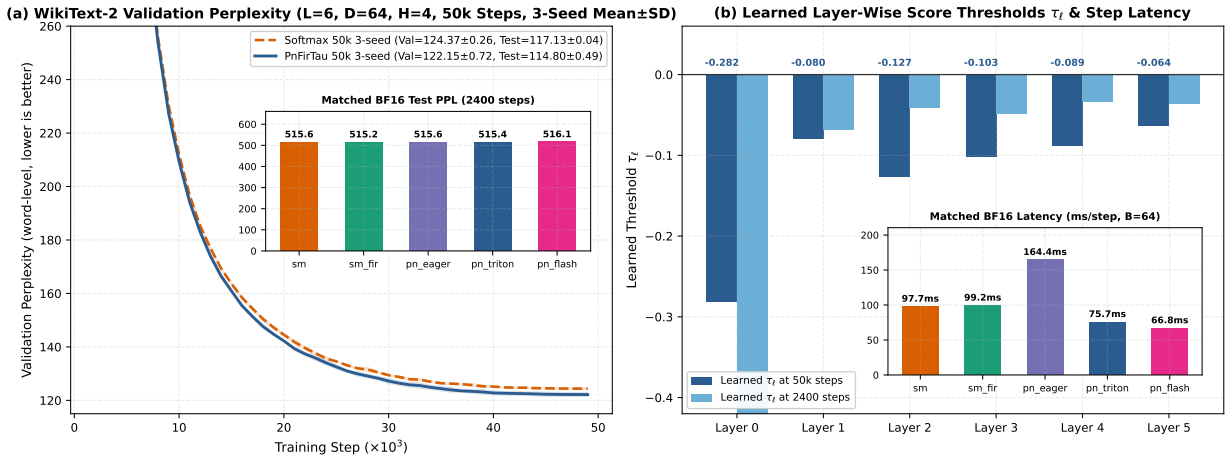


Figure 18: WikiText-2 50,000-step validation perplexity trajectory (3-seed mean±SD bands, suite 3 finals in legend) and 2,400-step matched BF16 inset (left), and learned layer-wise thresholds τ_ℓ with matched BF16 step-time inset across all 5 variants (right). Source: `results/exp4_wikitext2_lm.json`, `results/exp4d_softmax_50k_bf16.json`, `results/exp4e_multiseed_50k.json`.

only, so its resting point against pure-noise rows varies across seeds unless the post-warmup absent-key penalty (`pn_fir_tau_reg`) is applied.

6. **Hyperparameter sensitivity:** $p = 10$ and $\sigma = 2.0$ are fixed a priori; a sensitivity sweep over these values is left for future work.

5 Conclusion and Practical Guide for Transformer Designers

We have shown that causal FIR temporal smoothing strictly on K enables single-layer content-addressable value-slot binding ($W_{\text{query},j^*+1} = 0.729$–$0.792$, lifting clean recall from 35.8%–44.5% to $100.0\% \pm 0.0\%$ on Exp 1 and Exp 2), established mass constancy along uniform shift rays (Proposition 1), explained why uncalibrated `pn_fir` suffers score-shift starvation on positive-only training while simplex normalizers (`Softmax`, `Sparsemax`, `Entmax`) fail residual preservation on unanswerable queries (27.2%–34.4% hallucination rate in Exp 5) and row-mean centering (`pn_fir_c`) leaks $\approx 124\times$ more noise ($\sum W_{\text{unans}} = 0.265$), and demonstrated that `pn_fir_tau` achieves $100.0\% \pm 0.0\%$ residual preservation (0.0% hallucination rate, $\sum W_{\text{unans}} = 0.002 \pm 0.002$, $50\times$ lower continuous residual leakage than `Softmax_1` and `Quiet Softmax`) under pure Cross-Entropy loss while matching and slightly improving 6-layer WikiText-2 perplexity at 1.8M parameters.

For practitioners building and deploying Transformer models, our findings translate into two actionable architectural takeaways:

1. **Immediate Zero-Parameter Single-Layer Induction (`sm_fir`):** If the primary goal is to equip a standard causal Transformer with single-layer key-value induction binding (compressing two-layer induction heads into one layer) without custom attention kernels, applying the fixed 7-tap causal Gaussian FIR filter strictly to the key tensor K (`sm_fir`) requires zero learnable parameters, adds only +1.5% step-time overhead in `BF16` (+2.0% in `FP32`), and can be paired with `triton_causal_fir` for zero-copy strided filtering.

2. **Hallucination-Free Residual Preservation, Sink-Free Attention, and Transcendental-Free Hardware Acceleration (`pn_fir_tau_triton` / `pn_fir_tau_flash`):** If the architecture requires attention heads to **abstain cleanly when queried information is absent from the context**—such as RAG over noisy passages, memory-augmented Transformers, or models designed for low-bit INT4/INT8 quantization and streaming KV caches without `BOS` Attention-Sink outliers—replacing simplex normalizers with sub-simplex rectified power-normalization (`pn_fir_tau`) eliminates context-induced residual hallucinations (0.0% vs. 27.2%–34.4% for `Softmax`/`Sparsemax`/`Entmax`) and suppresses continuous absent-key residual leakage $50\times$ more tightly than virtual-sink Softmax. Moreover, when deployed via our fused Triton kernels (`PnFirTauTritonMHA` and `PnFirTauFlashMHA` in `src/triton_pnfirtau.py`), evaluating $(S'_{t,j})_+^{10}$ in 3 register multiplications with zero `exp()` calls and zero online tile rescaling reduces full-LM `BF16` step time by 22.5%–31.6% **below standard `Softmax`** ($2.17\times$ –$2.46\times$ faster than eager PyTorch) and achieves $21.70\times$ **speedup (**$6.97\times$ **vs. `softmax`) and** $66.5\times$ **VRAM reduction** on a 6-layer MHA stack at $N = 1024, B = 16$.

Declarations

Competing interests. The author declares no competing interests.

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Data and code availability. The reproducibility package (source code, experiment suites, raw JSON results, and figure-generation pipeline) is provided with this preprint; an archived copy will be deposited on Zenodo upon publication.

Author contributions. M.Z. conceived the study, implemented the method and kernels, ran the experiments, analysed the results, and wrote the manuscript.

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